



Semis Capex											
\$MM	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Memory Total	\$55.225	\$41.189	\$47.010	\$61.035	\$60.807	\$47.747	\$61.833	\$75.850	\$118.111	\$144.735	\$152.905
DRAM	\$25.053	\$17.816	\$23.790	\$31.252	\$35.248	\$36.283	\$47.121	\$60.522	\$95.238	\$113.602	\$116.613
NAND/Non-Volatile	\$30.172	\$23.373	\$23.221	\$29.783	\$25.559	\$11.463	\$14.712	\$15.328	\$22.873	\$31.133	\$36.292
Large non-Memory IDMs	\$14.692	\$18.960	\$16.384	\$22.312	\$33.734	\$35.754	\$30.778	\$23.655	\$24.957	\$30.043	\$42.640
Tier-1/Tier-2 Foundries	\$20.137	\$25.340	\$35.125	\$51.103	\$60.782	\$58.099	\$52.673	\$62.760	\$83.083	\$103.945	\$114.206
Other & SATS	\$16.746	\$17.044	\$19.049	\$28.713	\$32.016	\$32.170	\$30.188	\$36.786	\$43.000	\$48.136	\$52.721
Total	\$106.800	\$102.533	\$117.569	\$163.163	\$187.338	\$173.769	\$175.472	\$199.051	\$269.151	\$326.858	\$362.472

Capex YoY %											
Memory Total	00	00	00	00	00	00	00	00	01	00	00
DRAM	01	00	00	00	00	00	00	00	01	00	00
NAND/Non-Volatile	00	00	00	00	00	-01	00	00	00	00	00
Large non Memory IDMs	00	00	00	00	01	00	00	00	00	00	00
Tier-1/Tier-2 Foundries	00	00	00	00	00	00	00	00	00	00	00
Total	00	00	00	00	00	00	00	00	00	00	00

Segment % of Total											
Memory Total	01	00	00	00	00	00	00	00	00	00	00
DRAM	00	00	00	00	00	00	00	00	00	00	00
NAND/Non-Volatile	00	00	00	00	00	00	00	00	00	00	00
Logic	00	00	00	00	00	00	00	00	00	00	00
Foundries ex-DRAM	00	00	00	00	00	00	00	00	00	00	00

Semis Revenue, \$MM											
Memory Total	\$157.967	\$106.432	\$117.482	\$153.838	\$129.767	\$92.309	\$165.089	\$230.042	\$961.112	\$1.637.690	
% YoY	27,4%	-32,6%	10,4%	30,9%	-15,6%	-28,9%	78,8%	39,3%	317,8%	70,4%	
CAGR, trailing 5yy	18,7%	6,2%	8,8%	14,9%	0,9%	-10,2%	9,2%	14,4%	44,3%	66,0%	
DRAM	\$99.308	\$62.475	\$64.324	\$92.960	\$77.769	\$51.945	\$94.860	\$154.310	\$655.847	\$1.128.857	
NAND/Non-volatile	\$54.227	\$40.180	\$49.390	\$55.953	\$47.045	\$36.104	\$66.000	\$70.900	\$300.312	\$504.045	
Other	\$4.273	\$3.777	\$3.769	\$4.925	\$4.953	\$4.261	\$4.229	\$4.833	\$4.952	\$4.789	
Non-memory	\$310.812	\$305.670	\$322.907	\$402.054	\$442.617	\$429.164	\$437.085	\$515.424	\$663.017	\$740.361	
% YoY	7,9%	-1,7%	5,6%	24,5%	10,1%	-3,0%	1,8%	17,9%	28,6%	11,7%	
CAGR, trailing 5yy	5,4%	3,6%	4,6%	8,9%	9,0%	6,7%	7,4%	9,8%	10,5%	10,8%	
Total	\$468.778	\$412.101	\$440.389	\$555.893	\$572.384	\$521.473	\$602.173	\$745.467	\$1.624.128	\$2.378.051	
% YoY	00	00	00	00	00	00	00	00	01	00	

Capex Intensity %											
Memory Total	00	00	00	00	00	01	00	00	00	00	
DRAM	00	00	00	00	00	01	00	00	00	00	
NAND	01	01	00	01	01	00	00	00	00	00	
Non-memory	00	00	00	00	00	00	00	00	00	00	
Total	00	00	00	00	00	00	00	00	00	00	

WFE											
\$MM	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Memory Total	\$34.350	\$22.850	\$29.450	\$44.650	\$39.000	\$27.500	\$37.450	\$46.500	\$69.500	\$98.500	\$107.500
DRAM	\$15.550	\$10.250	\$13.400	\$23.100	\$22.500	\$20.750	\$29.400	\$34.750	\$52.000	\$74.000	\$78.000
NAND	\$18.800	\$12.600	\$16.050	\$21.550	\$16.500	\$6.750	\$8.050	\$11.750	\$17.500	\$24.500	\$29.500
Non-memory Total	\$19.500	\$28.050	\$30.550	\$45.850	\$61.550	\$62.900	\$64.050	\$69.500	\$77.500	\$99.500	\$140.000
Total	\$53.850	\$51.500	\$61.500	\$90.500	\$100.550	\$90.400	\$101.500	\$116.000	\$147.000	\$198.000	\$247.500
China WFE	\$6.000	\$7.000	\$14.000	\$17.000	\$22.800	\$30.500	\$39.000	\$44.500	\$47.500	\$56.000	\$64.000
% of total	11,1%	13,6%	22,8%	18,8%	22,7%	33,7%	38,4%	38,4%	32,3%	28,3%	25,9%

Multinationals in China	\$3.300	\$3.500	\$4.900	\$4.250	\$4.290	\$2.500	\$1.200	\$1.000	\$500	\$700	\$700	
Domestic China	\$2.700	\$3.500	\$9.100	\$12.750	\$18.510	\$28.000	\$37.800	\$43.500	\$47.000	\$55.300	\$63.300	
Memory	\$1.200	\$1.750	\$3.000	\$4.150	\$6.610	\$5.960	\$11.870	\$9.800	\$14.060	\$20.690	\$25.650	
Non-Memory	\$1.500	\$1.750	\$6.100	\$8.600	\$11.900	\$22.040	\$25.930	\$33.700	\$32.940	\$34.610	\$37.650	
Power/SiC/MEMS	\$0	\$0	\$0	\$258	\$384	\$492	\$632	\$824	\$986	\$1.168	\$1.300	
Non-Memory (ex-Power/SiC/MEMS) = Analog/M	\$1.500	\$1.750	\$6.100	\$8.342	\$11.516	\$21.548	\$25.298	\$32.875	\$31.954	\$33.442	\$36.351	
Total WFE ex-China	\$47.850	\$44.500	\$47.500	\$73.500	\$77.750	\$59.900	\$62.500	\$71.500	\$99.500	\$142.000	\$183.500	
Total WFE ex-China F/L & CXMT @ 50%	\$53.100	\$50.625	\$58.450	\$86.200	\$94.513	\$79.380	\$88.535	\$99.150	\$130.530	\$180.695	\$228.550	
WFE YoY %												
Memory Total	00	00	00	01	00	00	00	00	00	00	00	
DRAM	01	00	00	01	00	00	00	00	00	00	00	
NAND	00	00	00	00	00	-01	00	00	00	00	00	
Non-memory Total	00	00	00	01	00	00	00	00	00	00	00	
Total	00	00	00	00	00	00	00	00	00	00	00	
Segment % of total												
Memory Total	01	00	00	00	00	00	00	00	00	00	00	
DRAM	00	00	00	00	00	00	00	00	00	00	00	
NAND	00	00	00	00	00	00	00	00	00	00	00	
Non-memory Total	00	01	00	01	01	01	01	01	01	01	01	
Logic	00	00	00	00	00							
Foundry	00	00	00	00	00							
Total	01	01	01	01	01	01	01	01	01	01	01	
Domestic China Projects	00	00	00	00	00	00	00	00	00	00	00	
Total WFE ex-Domestic China	01	01	01	01	01	01	01	01	01	01	01	
Semis revenue, \$MM												
Memory Total	\$157.967	\$106.432	\$117.482	\$153.838	\$129.767	\$92.309	\$165.089	\$230.042	\$961.112	\$1.637.690		
DRAM	\$99.308	\$62.475	\$64.324	\$92.960	\$77.769	\$51.945	\$94.860	\$154.310	\$655.847	\$1.128.857		
NAND	\$54.227	\$40.180	\$49.390	\$55.953	\$47.045	\$36.104	\$66.000	\$70.900	\$300.312	\$504.045		
Other	\$4.273	\$3.777	\$3.769	\$4.925	\$4.953	\$4.261	\$4.229	\$4.833	\$4.952	\$4.789		
Non-memory	\$310.812	\$305.670	\$322.907	\$402.054	\$442.617	\$429.164	\$437.085	\$515.424	\$663.017	\$740.361		
Total	\$468.778	\$412.101	\$440.389	\$555.893	\$572.384	\$521.473	\$602.173	\$745.467	\$1.624.128	\$2.378.051		
% YoY	00	00	00	00	00	00	00	00	01	00		
WFE Intensity %												
Memory Total	00	00	00	00	00	00	00	00	00	00		
DRAM	00	00	00	00	00	00	00	00	00	00		
NAND	00	00	00	00	00	00	00	00	00	00		
Non-memory	00	00	00	00	00	00	00	00	00	00		
Total	00	00	00	00	00	00	00	00	00	00		
Ex-Dom China Mature F/L	00	00	00	00	00	00	00	00	00	00		
ex. Dom China Mature F/L & CXMT @ 50%	00	00	00	00	00	00	00	00	00	00		